LDACA ONLINE WORKSHOP · SESSION 2 OF 2

Coding text with GenAI: hands-on

The Annotation tool in LDaCA Wordflow · keyboards required, mistakes encouraged

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LDaCA | THE UNIVERSITY OF SYDNEY — Sydney Informatics Hub

ACKNOWLEDGEMENT OF COUNTRY

We acknowledge the **Gadigal people of the Eora Nation**, the Traditional Custodians of the land from which we present today, and the Traditional Custodians of the many lands from which you are joining.

We pay our respects to Elders past and present.

BEFORE WE START: 3 CHECKS

1. **Wordflow is running**: desktop app (v0.7.x) or Binder. Not yet? `sih.tools/wordflow`. Install takes ~5 minutes; follow on the shared screen meanwhile.
2. **Model access is provided in this session**: a shared workshop key lands in the chat when we need it (deleted at 3:30 pm). Your own OpenRouter / OpenAI / Anthropic / Google key works too, if you prefer.
3. **The hands-on sheet is open**: every step from this session, in order, with exact button names. Link in the chat now.

Just joining for the afternoon? Perfect. This session stands alone. The morning's recording covers the wider tour of Wordflow.

THE IDEA, HONESTLY

An LLM can code your text. Should it?

The promise

Classify thousands of texts against *your* categories in minutes: stance, topic, tone, anything you can define in words. No training data, no coding scripts.

The pitfalls

Confidently wrong labels · inconsistent on edge cases · biased where language is biased · drifts when the prompt is vague · and it never says "I'm not sure" unless you give it that option.

Today isn't "trust the AI"; it's **how to check it**. Every pitfall above has a control you'll use within the hour.

THE METHOD

Treat the AI like a new coder on your team.

1. **Codebook**: define the categories in writing, like instructions to a new research assistant.
2. **Pilot**: let it code a small sample first. Never the whole dataset on faith.
3. **Measure agreement**: compare against a reference: a human-verified annotation coded to the same codebook, or another coder. Intercoder reliability (Cohen's κ), confusion matrix.
4. **Revise**: sharpen the codebook and prompt where the disagreements pile up. Re-pilot.
5. **Document**: prompt, codebook, model, agreement scores → your methods section.

> A coder earns trust through agreement, human or machine.

BEFORE YOU DIVE IN

Using AI in research: your call, your responsibility.

Today we use shared **OpenRouter** access so everyone can do the exercise: convenience, **not an endorsement** of OpenRouter or any commercial AI service for research. A reminder, not a course.

Ethics & policies

Which providers and models you may use, and whether data may leave your machine at all, is set by your **ethics approval** plus your institution's, journal's and funder's AI rules; not by the tool.

Data privacy

Coding sends your texts to the provider. **Local models** keep data on your machine, but a model can still key on a name or an ethnicity: **de-identify regardless**, and consider synthetic data while developing.

Fit for the question

Match the model to your task, language, data **and methodology**: reflexive, interpretive analysis from a human lens has no model that fits. Pilot and measure; never assume.

Cost & accountability

Budget for full-corpus runs; document model, prompt, codebook and agreement. **You remain accountable** for everything the AI did on your behalf.

LDaCA · ARDC Australian Research Data Commons · Australian Government · NCRIS National Research Infrastructure for Australia · Sydney Corpus Lab · The University of Sydney — Sydney Informatics Hub

THE HANDS-ON, IN ONE SLIDE

1. **Build the Tweets block** (the morning's prep, brisk): workspace, sample data, column types, join with candidate metadata. Fell behind? Checkpoint a is exactly this result.
2. **Explore**: the Frequency word cloud says the campaign is about **jobs**. Two filters (drop retweets, keep "job") → everyone has the same **226 tweets**.
3. **Join the reference annotation**: human-verified labels for all 226 tweets, onto your block. Then the codebook: `Promise` / `Cuts` / `Other`; code the first page yourself and measure *your* agreement with the reference.
4. **Connect a model**: shared workshop key from the chat; a small, cheap model on purpose → new column `job.AI`.
5. **Preview → Compare To → confusion matrix**: measure the AI against your codes and the *reference annotation*; correct rows (reuse `job.manual`).
6. **Run All**: 226 tweets coded, κ against the reference. Then two more levers, each in its own column: your codes as **examples** (`job.AI.example`) and a **revised codebook** (`job.AI_v2`). Better? Measure, don't assume. Then the coded block goes into the other tools: that's where your analysis starts.

IF YOU FALL BEHIND

Checkpoints: rejoin in under a minute.

1. Download the latest checkpoint file from the link in the chat (a `.zip` workspace archive).
2. **Data Loader → Workspace manager → Upload workspace** → choose the file.
3. Click **Load** on the newly listed workspace.
4. Back in the **Annotation** tool: re-select the data block / column if a selector shows empty; everything else (data, tabs, results) is restored.

Your API key isn't touched: provider settings live on your machine, not in the workspace file. Use a checkpoint whenever it helps: they exist so you can spend your time on the interesting parts.

AFTER CODING AT SCALE

The coded block is data. Now ask it your question.

Coding was never the goal. A good codebook turns 226 tweets into a **variable**: your angle on the data, ready for every other tool in Wordflow.

Filter → a corpus per code

Preprocessing → Filter `job.AI_v2 = Cuts` → **Add to Workspace**: a block of just the cuts talk, ready for any tool below.

Frequency

Compare the `Promise` and `Cuts` vocabularies side by side: who says what when the subject is jobs.

Trends

Codes over the campaign, grouped by `party`: when did the cuts attacks peak, and from whom?

Concordance & Topic Modelling

Read one code in context; find the themes inside `Promise`. The code is a lens on your data, not the end of it.

Carry the provenance with it: codebook, prompt, model, κ. Every result downstream inherits them.

TAKE IT HOME

- **Today ran on a shared workshop key, which is now gone.** For real research: your own API key (the setup is identical to what you just did), or a local model: **Add Provider → Custom** points Wordflow at a local server (Ollama, LM Studio; anything speaking the OpenAI API), so your data never leaves your machine.
- **Check your ethics approval before coding real data with AI.** Which models, which providers, and whether data may be sent to an external API at all: that's your approval's call, not the tool's.
- **Keep the loop:** codebook → pilot → agreement → revise → document. Prompt, codebook, model and κ belong in your methods section.
- **No AI required: it's a manual coding tool too.** A team of human coders can each code their own column, and the same **Compare To** measures intercoder reliability between your target column and multiple reference columns: percent agreement, Cohen's κ, or Krippendorff's α, plus the confusion matrix.
- **Keep Wordflow**: `sih.tools/wordflow`. It's under active development: tap the in-app **Feedback** button whenever anything breaks or puzzles you. The app checks for new releases itself; updating to the latest version is one click. Cite via the sidebar's quote icon (“Cite LDaCA Wordflow”).

Data: Bruns, A.; Angus, D.; Cohen, T.; QUT Digital Observatory (2022). *Queensland Election 2020 on Twitter.* QUT. doi.org/10.25912/RDF_1665115527020 · Gender metadata: Sydney Corpus Lab.